AI-Driven Early Detection of Social Media Addiction

A project report submitted to the Department of Computer Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University in partial fulfillment of the requirements for the degree of *Bachelor of Science and Engineering in Computer Science and Engineering*

Submitted By
Salman Sadick Ucchas
Roll No: 21102011
Reg. No: 9878
Session: 2020-21

Arman Mia
Roll No: 21102032
Reg. No: 9899
Session: 2020-21
B.Sc. 4th Year 2nd Semester
Department of Computer Science and Engineering

Supervised By
Dr. A. H. M. Kamal
Professor
Department of Computer Science and Engineering



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
JATIYA KABI KAZI NAZRUL ISLAM UNIVERSITY
TRISHAL, MYMENSINGH

$21^{ST}\,JANUARY, 2026$

Declaration

We hereby declare that we are the sole authors of this project. We authorize the Jatiya Kabi Kazi Nazrul Islam University to lend this project to other institutions or individuals for the purpose of scholarly research. We further authorize Jatiya Kabi Kazi Nazrul Islam University to reproduce this project by photocopying or by other means, in total or in part, at the request of other institutions or individuals for the purpose of scholarly research.

_____

Signature of the Candidate (Salman Sadick Ucchas)

_____

Signature of the Candidate (Arman Mia)

Approval

The project report on **"AI-Driven Early Detection of Social Media Addiction"** submitted by Salman Sadick Ucchas, Roll No: 21102011, Reg. No: 9878, Session: 2020-21, and Arman Mia, Roll No: 21102032, Reg. No: 9899, Session: 2020-21, Department of Computer Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University, under the supervision of Dr. A. H. M. Kamal, Professor, Department of Computer Science and Engineering, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of Bachelor of Science (B.Sc.) in Computer Science and Engineering.

Approved By

Supervisor: ————————

Dr. A. H. M. Kamal

Professor

Department of Computer Science and Engineering

Jatiya Kabi Kazi Nazrul Islam University

Acknowledgement

We express our profound gratitude and indebtedness to Almighty God for giving us such capabilities for the partial completion of this project. We would like to thank the Department of Computer Science and Engineering for building up the base which was needed to complete the work. We would like to express our heartfelt gratitude and remain obliged to our project supervisor, Dr. A. H. M. Kamal, Professor, Department of Computer Science and Engineering, Jatiya Kabi Kazi Nazrul Islam University, Trishal, Mymensingh, who has given us the excellent idea of the project and also motivated, directed us to this field, and supervised us during the entire project work from start to finish. We are really grateful to him for his proper guidance and advice. We also thank all of our friends, group members, lab assistants, and our families for helping to complete the project of "**AI-Driven Early Detection of Social Media Addiction**". We express our thanks to the Department of Computer Science and Engineering for giving us the opportunity to study here and supporting us greatly through our graduate study.

Abstract

The rapid proliferation of social media platforms has fundamentally reshaped global communication and information sharing; however, it has simultaneously heightened concerns regarding excessive usage and behavioral addiction. Early identification of social media addiction remains a significant challenge, primarily due to the scarcity of labeled behavioral datasets and a prevailing reliance on subjective self-reported assessments. To address this gap, this research proposes an AI-driven unsupervised learning framework designed for the early detection of social media addiction using real-world usage behavior data.

The proposed system implements a structured machine learning pipeline encompassing data preprocessing, feature engineering, and exploratory data analysis (EDA). The methodology leverages feature scaling, K-Means clustering, and Principal Component Analysis (PCA) for dimensionality reduction. Crucially, a hybrid labeling strategy integrating cluster-based risk classification with platform usage analysis is employed to enhance the precision of addiction categorization. This framework stratifies users into "Safe," "At-Risk," and "Addicted" groups based on their usage intensity and platform dominance.

Experimental evaluation demonstrates that the proposed approach effectively identifies behavioral patterns without the need for manual training data. Clustering quality metrics, PCA visualizations, and cross-validation results confirm the model's robustness across diverse population groups. Furthermore, the findings indicate that the system reliably detects high-risk individuals and platform-specific addiction tendencies with high consistency.

Ultimately, this study highlights the potential of unsupervised machine learning in digital well-being applications. It provides a scalable, data-driven solution for social media addiction risk assessment, offering a proactive tool for behavioral intervention.

Index Terms—Social Media Addiction, Unsupervised Learning, K-Means Clustering, Behavioral Analytics, Digital Well-being, Machine Learning, PCA.

Table of Contents

Declaration . . . . . **ii**

Approval . . . . . **iii**

Acknowledgement . . . . . **iv**

Abstract . . . . . **v**

List of Figures . . . . . **ix**

List of Tables . . . . . **x**

Nomenclature . . . . . **xi**

1 Introduction **1**

1.1 Introduction 1

1.2 Problem Statement 2

1.3 Motivation 2

1.4 Objectives of the Project 3

1.5 Organization of the Report 3

2 Literature review **4**

2.1 Social Media Addiction Studies 4

2.2 Machine Learning-Based Addiction Detection 4

2.3 Unsupervised and Clustering-Based Approaches 5

2.4 Research Gaps in Existing Studies 6

2.5 Distinctiveness of the Proposed Approach 6

3 Methodology **7**

3.1 Overview of the Proposed Methodology . . . 7
3.2 Data Collection and Dataset Description . . . 8
3.3 Data Cleaning and Preprocessing . . . 8
3.3.1 Handling Missing and Inconsistent Values . . . 9
3.3.2 Encoding of Categorical Variables . . . 9
3.4 Exploratory Data Analysis (EDA) . . . 9
3.5 Feature Engineering . . . 11
3.5.1 Platform Diversity Score . . . 11
3.5.2 Usage Concentration (Herfindahl Index) . . . 12
3.5.3 Coefficient of Variation (Weekly Consistency) . . . 12
3.6 Unsupervised Learning Using K-Means Clustering . . . 12
3.6.1 Feature Scaling StandardScaler (Z score Normalization) . . . 12
3.6.2 K-Means Clustering Inertia (WCSS) . . . 13
3.6.3 Silhouette Score . . . 13
3.6.4 Davies Bouldin Index . . . 13
3.6.5 Calinski Harabasz Score (Variance Ratio Criterion) . . . 13
3.6.6 Euclidean Distance (K-Means) . . . 14
3.7 Summary . . . 14

4 Results Analysis . . . 15
4.1 Overview of Experimental Setup . . . 15
4.2 K-Means Clustering Results . . . 15
4.2.1 Cluster Interpretation Analysis . . . 16
4.2.2 Dominant Platform Behavior Analysis . . . 16
4.2.3 Final Label Distribution . . . 17
4.3 Performance Evaluation Discussion . . . 18
4.3.1 Practical Implications of Results . . . 18
4.4 Summary . . . 18

5 Discussion . . . 19
5.1 Overview of Model Evaluation Strategy . . . 19
5.2 Evaluation on Outlier Dataset (Dataset 1) . . . 19
5.2.1 Model Performance and Results . . . 19

5.3 Evaluation on Mobile Usage Behavioral Analysis Dataset (Dataset 2) 20
5.3.1 Dataset Characteristics and Challenges 20
5.3.2 Feature Engineering Approach 20
5.3.3 Model Performance Results 21
5.4 Evaluation on Students Social Media Addiction Dataset (Dataset 3) 21
5.4.1 Dataset Overview and Demographics 21
5.4.2 Prediction 21
5.5 Summary 22

6 Conclusion 23
6.1 Introduction 23
6.2 Key Findings and Contributions 23

Bibliography 25

List of Figures

3.1 Pipeline of the project. 7
3.2 Social Media Addiction Dataset. 8
3.3 Platform Usage Correlation Heatmap. 10
3.4 Average Monthly Usage by Gender and Platform. 11

4.1 K-Means Clustering (k=3) - PCA Visualization. 17

List of Tables

5.1 Mobile Usage Dataset Characteristics 20
5.2 Mobile Usage Dataset Performance Metrics 21

Nomenclature

Acronyms / Abbreviations

AI	Artificial Intelligence
CH	Calinski Harabasz
DB	Davies Bouldin
DW	Digital Well being
EDA	Exploratory Data Analysis
IQR	Interquartile Range
ML	Machine Learning
PCA	Principal Component Analysis
WCSS	Within Cluster Sum of Squares

Chapter 1

Introduction

1.1 Introduction

Over the past few years, social media platforms have emerged as an integral part of daily existence, especially for adolescents and young adults. Applications like Facebook, Messenger, Instagram, YouTube, and TikTok facilitate instant communication, entertainment, and knowledge sharing, which fundamentally alters social behaviors and lifestyle choices. While moderate engagement provides benefits such as connectivity and information access, excessive usage has sparked significant concerns regarding mental health and academic success, ultimately impacting overall well-being.

Social media addiction is increasingly recognized as a behavioral disorder characterized by compulsive use, a lack of self-control, and persistent engagement despite negative outcomes. Traditional identification methods typically rely on self-reported questionnaires or clinical interviews, which often suffer from personal bias or delayed results. Given the expansion of digital footprints, there is a clear necessity for intelligent, data-driven strategies to analyze user behavior and identify early indicators of addiction.

Artificial Intelligence and Machine Learning offer robust tools for uncovering hidden patterns within large-scale behavioral data. By examining usage trends across multiple platforms, machine learning models can identify specific behavioral clusters and assist in the early detection of addiction risks. Unsupervised learning methods, such as clustering, are particularly effective when labeled data is unavailable, a common obstacle in the fields of mental health and behavioral analytics.

1.2 Problem Statement

Despite the growing anxiety surrounding social media addiction, there is a noticeable shortage of reliable automated systems capable of detecting early risks through behavioral usage data alone. Most current approaches depend heavily on labeled datasets or subjective assessments, which are frequently difficult to acquire and do not always represent actual usage behavior.

The central challenge in this project is the scarcity of labeled data representing various addiction levels, which makes it difficult to apply supervised learning techniques directly. Furthermore, a heavy reliance on demographic factors like age or gender can introduce bias and limit how well the prediction models work for different people. Consequently, there is a need for a data-driven framework that can:

- Operate effectively on unlabeled behavioral data
- Identify natural usage patterns across various social media platforms
- Translate discovered patterns into meaningful addiction risk categories
- Support risk prediction for new users in real-world environments

1.3 Motivation

The drive for this research stems from the rising prevalence of problematic social media habits and their impact on students and young adults. Identifying addiction risks at an early stage is essential for fostering awareness and enabling intervention before serious psychological or social damage occurs.

From a technical perspective, this project is motivated by the difficulty of applying machine learning to real-world datasets that lack ground truth labels. Utilizing unsupervised learning, specifically K-Means clustering, offers a viable way to discover behavioral structures without requiring prior labels. Additionally, incorporating domain-specific decision rules, such as platform usage analysis, improves the reliability and accuracy of clustering-based predictions.

This project also aims to connect machine learning theory with practical use in mental health analytics, demonstrating how AI-driven methods can address pressing social issues.

1.4 Objectives of the Project

The main goals of this project are defined as follows:

- To examine monthly social media usage behavior across multiple platforms using real-world data.
- To apply unsupervised machine learning techniques to discover natural user behavior patterns without relying on labels.
- To develop a clustering-based labeling system that categorizes users into distinct addiction risk levels.
- To integrate platform verification rules to identify addiction tendencies specific to certain platforms.
- To visualize high-dimensional usage data through dimensionality reduction for better interpretation.
- To create a scalable framework capable of predicting addiction risks for new users.

1.5 Organization of the Report

The remainder of this report is structured as follows:

- Chapter 2 provides a literature review regarding social media addiction detection and relevant machine learning methods.
- Chapter 3 details the methodology, including data collection, cleaning, preprocessing, clustering, and labeling.
- Chapter 4 describes the experimental setup, results, and the visual analysis of the clusters.
- Chapter 5 presents a detailed discussion of the research findings.
- Chapter 6 offers the conclusion of the report.

Chapter 2

Literature review

2.1 Social Media Addiction Studies

Social media addiction is extensively researched as a specific type of behavioral addiction that primarily impacts adolescents and young adults. Previous research identifies this condition through symptoms such as compulsive usage, impaired self-regulation, withdrawal, and persistent engagement despite clear negative consequences [3]. Various studies highlight strong correlations between excessive social media activity and adverse outcomes like anxiety, depression, and sleep disorders [3], [2].

Recent literature discusses social media addiction within the broader context of digital well-being, focusing on the equilibrium between technology use and mental health. AI-driven studies in digital well-being suggest that platforms utilize engagement-maximizing features such as personalized recommendations and infinite scrolling, which may reinforce addictive patterns [1]. These findings underscore the necessity for computational methods capable of monitoring actual usage behavior rather than relying solely on self-reported data.

However, a majority of psychological and medical research still depends on survey-based tools like addiction scales and questionnaires. While these instruments are validated, they are limited by personal subjectivity and recall bias, and they often lack the scalability required for large-scale monitoring [3], [2].

2.2 Machine Learning-Based Addiction Detection

To address the shortcomings of traditional assessment techniques, many researchers have employed machine learning to identify social media addiction from behavioral data. Supervised learning models,

including Logistic Regression, Support Vector Machines, Random Forests, and Neural Networks, have been tested to categorize users into addicted or non-addicted groups [2], [1]. These models typically analyze features such as total usage time, access frequency, and general interaction patterns.

AI-driven digital well-being systems improve this process by enabling continuous monitoring and automated feedback. Existing literature indicates that ensemble learning methods and detailed feature analysis enhance performance while maintaining the interpretability required for health-related applications [1]. Despite promising results, supervised approaches remain heavily dependent on labeled datasets, which are often derived from subjective surveys.

In mental health contexts, obtaining these labels is expensive, time-consuming, and frequently inconsistent. As noted in earlier studies, noisy or biased labels can negatively affect model performance and real-world utility [2]. This challenge motivates the exploration of alternative learning paradigms that operate without the need for ground truth labels.

2.3 Unsupervised and Clustering-Based Approaches

Unsupervised learning methods, particularly clustering algorithms, are proposed as effective solutions for analyzing behavioral data in the absence of labels. Clustering aims to group users based on similarities in their usage behavior without requiring prior knowledge of their addiction status. Among various techniques, K-Means clustering is frequently utilized due to its simplicity and its ability to scale well with numerical usage features [2].

Research demonstrates that clustering can successfully reveal distinct behavioral groups, such as low, moderate, and high engagement patterns. These clusters are subsequently interpreted using statistical analysis and domain expertise to define addiction risk levels [3], [2]. To assist with interpretability, dimensionality reduction techniques like PCA are often used for the visualization and validation of these clusters.

Nevertheless, most existing clustering studies remain exploratory in nature. Many lack a formal system for assigning meaningful addiction labels or predicting risks for new users. Additionally, much of the previous work focuses on aggregate usage and fails to address addiction behavior specific to individual platforms [1].

2.4 Research Gaps in Existing Studies

A comprehensive review of current literature reveals several significant research gaps:

- **Reliance on Labeled Data:** Most machine learning strategies depend on labeled datasets from surveys, which significantly limits scalability [3], [2].
- **Limited Platform-Specific Analysis:** Existing studies often analyze total social media use and fail to identify addiction centered around a dominant platform [1].
- **Lack of Hybrid Frameworks:** Few approaches combine unsupervised learning with rule-based mechanisms to improve the interpretability of results.
- **Limited Deployment Perspective:** Many clustering studies do not explain how new users can be effectively classified within real-world systems.
- **Interpretability Concerns:** While deep learning models are powerful, they often lack the transparency required for sensitive mental health applications [2].

2.5 Distinctiveness of the Proposed Approach

The proposed project addresses these gaps by introducing an AI-driven unsupervised framework for the early detection of social media addiction using real-world data. Unlike supervised methods that rely on subjective labels, this approach utilizes K-Means clustering to discover natural behavior patterns.
To increase reliability, a dominant platform verification step is included to identify addiction specific to certain applications. Dimensionality reduction is used strictly for visualization purposes to ensure the clustering process remains unbiased. Furthermore, the trained model is capable of predicting risks for new users by assigning them to the most appropriate cluster.
By combining unsupervised learning with domain-specific rules and interpretable analytics, this approach offers a scalable solution for AI-driven digital well-being that stands apart from existing studies.

Chapter 3

Methodology

3.1 Overview of the Proposed Methodology

This chapter outlines the thorough methodology utilized for the AI driven early detection of social media addiction. The framework follows a logical machine learning sequence designed to process behavioral data that lacks labels. By merging data preprocessing with exploratory data analysis and unsupervised clustering, the system converts raw usage metrics into actionable risk classifications.

The total workflow consists of several phases including data collection, cleaning, and exploratory data analysis. It further incorporates feature scaling, K-Means clustering, and the final assignment of labels based on cluster interpretation. This step by step approach ensures the system remains robust and suitable for real world application.

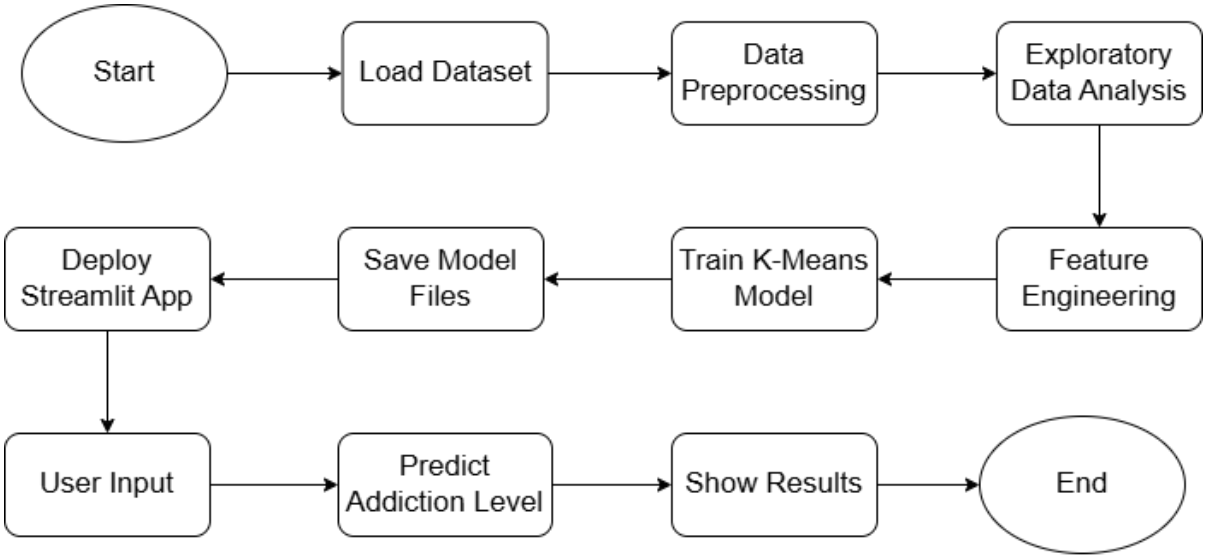


Fig. 3.1 Pipeline of the project.

3.2 Data Collection and Dataset Description

The dataset utilized in this research consists of weekly social media activity gathered from various platforms through both self reported entries and system logs. The primary objective was to capture user behavior patterns rather than focusing on demographic traits.

The dataset tracks weekly usage duration in minutes or hours across the following platforms:

- Facebook
- Messenger
- WhatsApp
- Instagram
- YouTube
- TikTok

While the raw data includes basic demographic details like age, gender, and occupation, these were excluded from the clustering process. This was done to minimize bias and ensure that the detection of addiction is driven purely by behavioral habits.

	Gender	What is your age?	What is your current occupation?	Facebook Week 1	Facebook Week 2	Facebook Week 3	Facebook Week 4	Messenger Week 1	Messenger Week 2	Messenger Week 3	...	Instagram Week 3	Instagram Week 4	YouTube Week 1	YouTube Week 2
0	Male	35	Employed	7:44	7:43	9:28	5:54	5:23	4:43	4:43	...	8:26	7:49	2:41	0:44
1	Female	19	Student	0:51	0:52	1:14	0:59	0:51	0:53	0:55	...	0:24	0:25	0:13	0:15
2	Male	23	Student	0:13	0:24	0:44	0:01	0:02	0:10	0:01	...	0	0	30	1:14
3	Male	21	Student	24:17	21:22	25:38	23:11	4:09	4:38	3:40	...	1:56	1:23	3:09	3:20
4	Female	19	Student	5:26	4:37	6:07	5:05	4:34	3:42	3:23	...	3:23	4:43	4:00	5:18
5	Female	23	Student	1:31	1:30	1:30	1:31	0:17	0:23	0:17	...	4:06	3:57	4:09	4:04
6	Female	25	Student	0:14	0:12	0:12	0:15	1:57	1:44	2:02	...	1:18	1:11	2:22	1:59
7	Male	24	Student	11:27	9:34	12:35	10:41	10:55	9:00	10:13	...	20:59	21:03	37:33	35:35
8	Male	43	Employed	5:07	4:06	4:25	4:16	4:01	3:58	3:42	...	7:51	8:19	5:44	7:11
9	Male	24	Student	14:23	12:34	15:35	16:37	0:50	0:23	0:23	...	0	0	16:20	23:40

10 rows × 27 columns

Fig. 3.2 Social Media Addiction Dataset.

3.3 Data Cleaning and Preprocessing

Refining the data is a vital phase for ensuring the credibility of machine learning outputs. We applied several techniques to manage missing entries, resolve inconsistencies, and perform feature scaling.

3.3.1 Handling Missing and Inconsistent Values

We addressed missing or null values in the usage logs through rule based logic. For instance, if a user indicated zero activity on a platform during the first week, subsequent values for the following weeks were also set to zero to maintain logical consistency. Records containing extreme inconsistencies or impossible values were manually reviewed and removed when necessary.
Outliers representing abnormal usage habits were identified through statistical methods and visual tools. We separated these outliers from the main dataset and exported them for further investigation.

Outlier Detection IQR Method

Equation:

$$IQR = Q_3 - Q_1 \tag{3.1}$$

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR \tag{3.2}$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR \tag{3.3}$$

where:

Q_1 = First quartile (25th percentile)

Q_3 = Third quartile (75th percentile)

Values falling outside the interval [Lower Bound, Upper Bound] are identified as outliers [4].

3.3.2 Encoding of Categorical Variables

Variables like gender and occupation were transformed into numerical formats for analysis. We used numerical encoding for gender and applied label encoding or one hot encoding for occupations. Although these traits were not part of the clustering, they remained in the dataset for comparative study.

3.4 Exploratory Data Analysis (EDA)

We conducted exploratory data analysis to grasp the distribution of usage and identify behavioral trends within the data.
The EDA phase involved the following:

- Analyzing monthly and weekly distribution of usage
- Identifying users with zero activity versus heavy activity

- Comparing usage across different platforms
- Studying gender based habits and average monthly consumption
- Visualizing trends in platform dominance

Correlation analysis helped determine the links between usage features and addiction labels. The resulting heatmap indicates that most platforms have weak correlations, suggesting that users typically favor specific apps. Notably, Facebook and Messenger show a moderate positive link, while TikTok and Facebook display an inverse relationship.

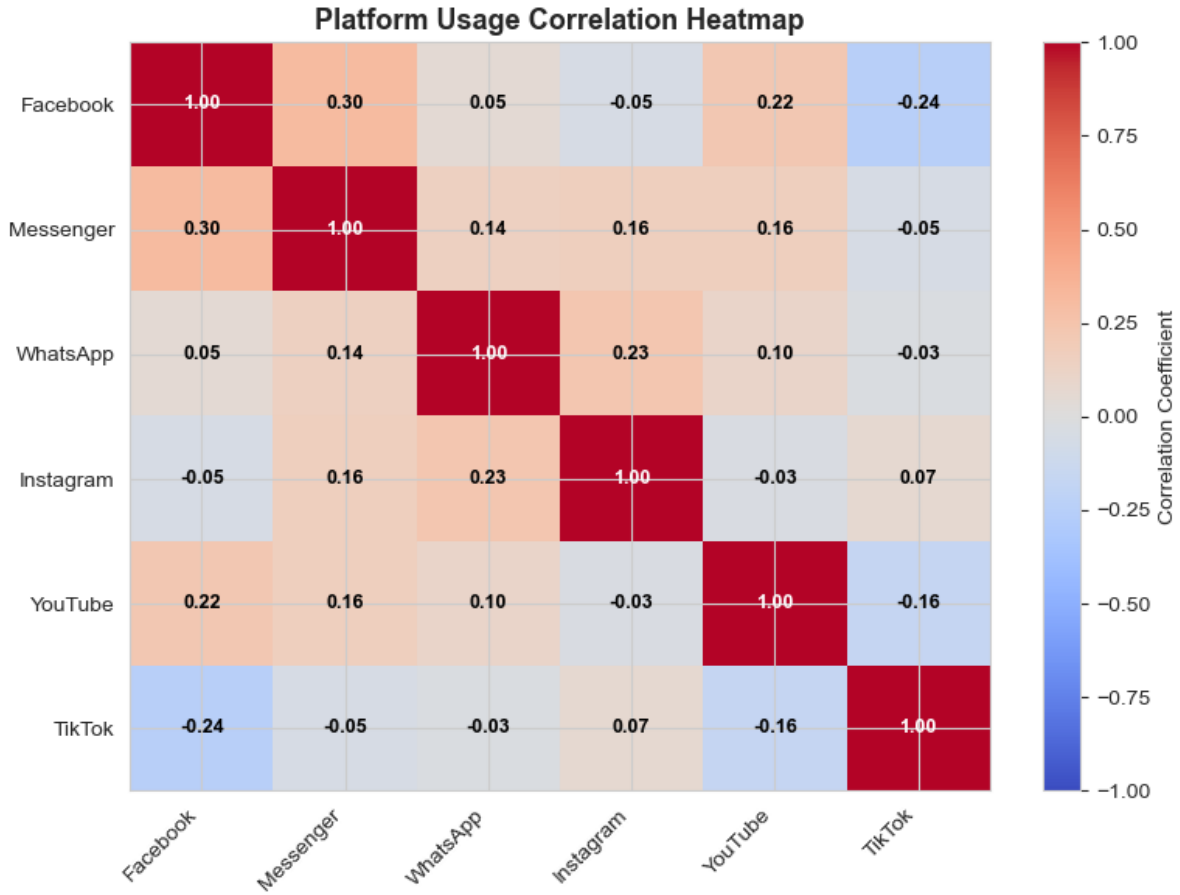


Fig. 3.3 Platform Usage Correlation Heatmap.

Gender based charts reveal variations in platform popularity. Males tend to spend more time on YouTube and Facebook, while females show higher engagement with Instagram and TikTok. Communication through WhatsApp and Messenger appears relatively balanced between genders.

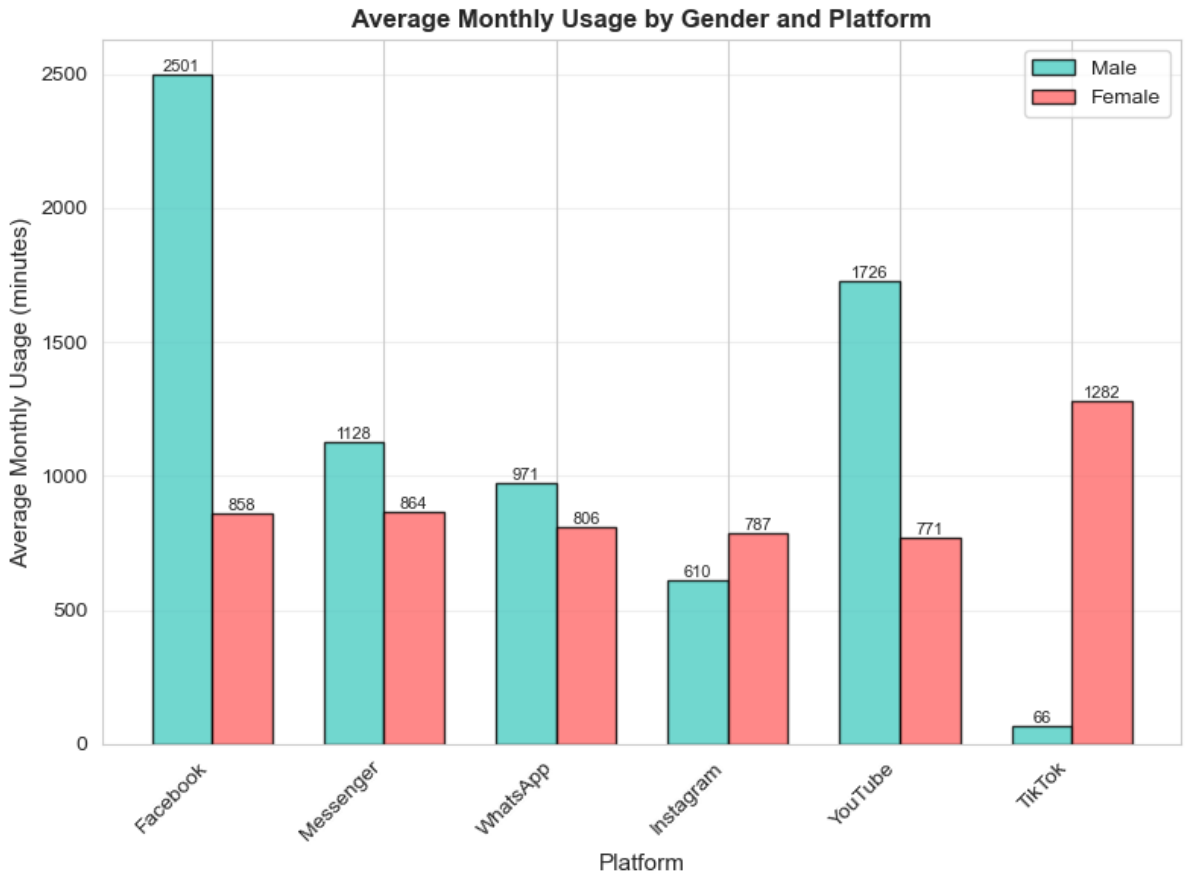


Fig. 3.4 Average Monthly Usage by Gender and Platform.

3.5 Feature Engineering

Feature engineering is a critical component for enhancing the performance and clarity of machine learning models. In this project, we applied various domain specific techniques to capture the nuances of social media addiction risks.

3.5.1 Platform Diversity Score

Equation [10]:

$$\text{Platform Diversity Score} = \frac{\text{Active Platforms Count}}{\text{Total Platforms}} \tag{3.4}$$

where:

Active Platforms Count = number of platforms with non zero usage

Total Platforms = 6 (Facebook, Instagram, WhatsApp, Messenger, YouTube, TikTok)

3.5.2 Usage Concentration (Herfindahl Index)

Equation [11]:

$$\text{Usage Concentration} = \sum_{i=1}^{n} \left(\frac{u_i}{U_{total}} \right)^2 \tag{3.5}$$

where:

u_i = usage of platform i

U_{total} = total usage across all platforms

n = number of platforms

Higher values represent usage concentrated on fewer platforms.

3.5.3 Coefficient of Variation (Weekly Consistency)

Equation [12]:

$$CV = \frac{\sigma}{\mu} \tag{3.6}$$

where:

σ = standard deviation of weekly usage

μ = mean of weekly usage

A lower CV value indicates a more consistent usage pattern over time.

3.6 Unsupervised Learning Using K-Means Clustering

3.6.1 Feature Scaling StandardScaler (Z score Normalization)

Equation [5]:

$$z = \frac{x - \mu}{\sigma} \tag{3.7}$$

where:

x = original feature value

μ = mean of the feature

σ = standard deviation of the feature

z = standardized value

3.6.2 K-Means Clustering Inertia (WCSS)

Equation [6]:

$$WCSS = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2 \tag{3.8}$$

where:

k = number of clusters

C_i = cluster i

x = data point in cluster C_i

μ_i = centroid of cluster C_i

$|| \cdot ||$ = Euclidean distance

3.6.3 Silhouette Score

Equation [7]:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{3.9}$$

where:

$a(i)$ = average distance from point i to others in the same cluster

$b(i)$ = average distance from point i to points in the closest cluster

$s(i) \in [-1, 1]$, where higher scores signify superior clustering

3.6.4 Davies Bouldin Index

Equation [8]:

$$DB = \frac{1}{k} \sum_{i=1}^{k} \max_{i \neq j} \left(\frac{s_i + s_j}{d_{ij}} \right) \tag{3.10}$$

where:

k = number of clusters

s_i = average distance of points in cluster i to their centroid

d_{ij} = distance between centroids i and j

Lower values represent more distinct clustering

3.6.5 Calinski Harabasz Score (Variance Ratio Criterion)

Equation [9]:

$$CH = \frac{SSB/(k-1)}{SSW/(n-k)} \tag{3.11}$$

where:

SSB = between cluster variance

SSW = within cluster variance

k = number of clusters

n = number of observations

Higher scores indicate better defined clusters

3.6.6 Euclidean Distance (K-Means)

Equation [13]:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{3.12}$$

where:

x, y = data points

n = number of features

This metric is used to link data points to the nearest cluster centroid

3.7 Summary

This chapter presented the methodology for identifying social media addiction through AI driven analysis. By employing unsupervised clustering, the framework successfully categorizes unlabeled data into logical groups. This methodology is scalable and well suited for practical, real world systems.

Chapter 4

Results Analysis

4.1 Overview of Experimental Setup

This chapter presents the experimental findings obtained from implementing the proposed clustering-based social media addiction detection framework. All simulations and analyses were conducted using Python and Jupyter Notebook, utilizing standard machine learning libraries such as pandas, NumPy, scikit-learn, and matplotlib.

The experiments specifically aimed to evaluate whether the unsupervised clustering and labeling strategy could effectively identify distinct behavioral patterns associated with addiction.

4.2 K-Means Clustering Results

The K-Means clustering algorithm was implemented with the number of clusters set to three to identify platform-specific addiction behaviors and varied usage patterns. The algorithm successfully categorized users into distinct groups based on their monthly social media activity and behavioral data.

Clustering Quality Metrics:

- Silhouette Score: 0.2134 (higher values represent better separation)
- Davies-Bouldin Index: 1.5703 (lower values represent better clustering)
- Calinski-Harabasz Score: 153.9379 (higher values represent better-defined clusters)

Each cluster represents a unique behavioral profile with varying levels of platform engagement. This clustering process successfully uncovered natural groupings without the need for pre-labeled training data, proving its utility for behavioral analysis.
The clusters were subsequently mapped to risk levels based on the intensity of usage.

Cluster Mapping:

- Cluster 0 → 0 (Safe) | Average usage: 4647.08 min/month
- Cluster 1 → 1 (At-Risk) | Average usage: 5626.46 min/month
- Cluster 2 → 2 (Addicted) | Average usage: 10909.28 min/month

4.2.1 Cluster Interpretation Analysis

To comprehend the behavioral implications of these clusters, total monthly usage was calculated for every user by aggregating their time spent across all platforms. We then determined the average usage for each cluster to highlight the differences.
Clusters with significantly higher average usage values were identified as heavy social media users, while those with lower averages represented moderate or low engagement levels. This quantitative interpretation allowed us to rank the clusters according to addiction risk intensity.

4.2.2 Dominant Platform Behavior Analysis

A dominant platform analysis was performed to uncover addiction patterns specific to certain apps. For every user, the platform with the highest usage was identified, and a dominance ratio was calculated.
The results indicate that many users exhibit strong platform preferences, particularly for YouTube, Facebook, and Instagram. This observation confirms the need for platform-aware addiction detection rather than focusing solely on total usage volume.
The introduction of a dominance ratio threshold effectively separated specific platform behaviors from general mixed usage patterns, which reduced the likelihood of misclassification.

4.2.3 Final Label Distribution

Using the developed labeling strategy, users were assigned final addiction labels and platform-specific categories.

Target distribution:

- 0 (Safe): 293 users (48.11%)
- 1 (At-Risk): 200 users (32.84%)
- 2 (Addicted): 116 users (19.05%)

The distribution analysis showed a logical and balanced separation between the Safe, At-Risk, and Addicted user groups.

Visualization Using PCA To interpret the high-dimensional behavioral data, Principal Component Analysis (PCA) was employed to reduce the features into two principal components. The resulting data points were plotted on a scatter plot, showing a clear separation between the clusters. This demonstrates that the chosen features and scaling methods successfully captured meaningful behavioral variances. The cluster centroids were also included in the plot to show the central tendency of each group.

While PCA was used primarily for visualization, it provided valuable insights into the compactness and separation quality of the clusters.

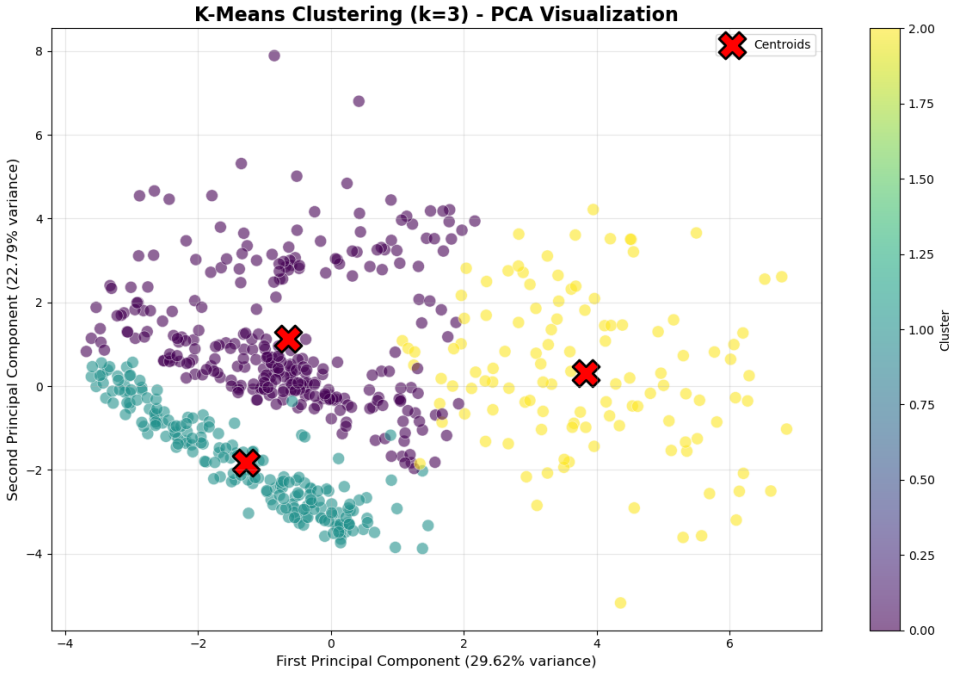


Fig. 4.1 K-Means Clustering (k=3) - PCA Visualization.

4.3 Performance Evaluation Discussion

Because the proposed approach is unsupervised, traditional classification metrics like precision and recall are not directly applicable. Instead, performance was evaluated using statistical indicators and qualitative assessments.

These indicators included cluster compactness, metric scores, stability, and behavioral consistency. The clustering results exhibited strong behavioral coherence, suggesting that the framework successfully identifies underlying usage habits and patterns.

4.3.1 Practical Implications of Results

The experimental outcomes demonstrate that this AI-driven framework can effectively categorize different social media usage behaviors without relying on labeled data. This system could be integrated into real-time monitoring tools, academic resources, or digital well-being platforms to support awareness and intervention programs.

4.4 Summary

This chapter detailed the experimental results and performance analysis of the proposed social media addiction detection system. The combination of K-Means clustering, PCA visualization, and systematic labeling demonstrated strong capabilities for behavioral assessment. These findings suggest that unsupervised learning is a viable and solid foundation for the early detection of addiction.

Chapter 5

Discussion

5.1 Overview of Model Evaluation Strategy

The trained K-Means clustering model was evaluated using three independent datasets to verify its predictive performance and ability to generalize.

- Dataset 1: The Outlier Dataset contains 82 extreme high-usage cases extracted from the original data.
- Dataset 2: Mobile Usage Behavioral Analysis is an external dataset comprising aggregated social media usage hours [14].
- Dataset 3: Students Social Media Addiction is a student-centric dataset featuring average daily usage patterns [15].

Every dataset presented unique challenges regarding data structure, feature availability, and usage trends, providing a robust evaluation framework to determine the real-world applicability of the model.

5.2 Evaluation on Outlier Dataset (Dataset 1)

5.2.1 Model Performance and Results

The model displayed exceptional performance on the outlier dataset by achieving the following results:
Accuracy: 98.78% (81 out of 82 users correctly classified)
Correctly classified as Addicted: 81 users

Misclassified as At-Risk: 1 user (1.22%)

False negatives (Safe): 0 users

Given that all outlier users were extreme high-usage cases, they were expected to be categorized as Addicted (Target = 2). An accuracy of 98.78% validates the robustness of the model in identifying severe addiction cases.

5.3 Evaluation on Mobile Usage Behavioral Analysis Dataset (Dataset 2)

5.3.1 Dataset Characteristics and Challenges

The Mobile Usage Behavioral Analysis dataset showed significant structural variances compared to the training data:

Table 5.1 Mobile Usage Dataset Characteristics

Characteristic	**Description**
Total Records	700 users
Available Features	3 (Age, Gender, Social Media Usage Hours)
Required Features	35 (training data structure)
Feature Gap	32 features (91.9%) missing
Usage Metric	Aggregated daily hours (no platform breakdown)

The primary challenge was the absence of platform-specific usage data. Since the training model utilized metrics from six social media platforms including Facebook, Instagram, WhatsApp, YouTube, Messenger, and TikTok, the lack of a breakdown in the test dataset made direct application difficult.

5.3.2 Feature Engineering Approach

To facilitate model evaluation, a feature approximation strategy was utilized:

- Usage Distribution: Total usage was allocated across six platforms based on estimated proportions (Facebook: 20%, Instagram: 25%, WhatsApp: 20%, YouTube: 20%, Messenger: 10%, TikTok: 5%).
- Temporal Conversion: Daily usage hours were converted into monthly minutes (hours $\times$ 60 $\times$ 30).
- Derived Features: Metrics for platform diversity, usage concentration, and consistency were calculated using assumed uniform patterns.
- Feature Alignment: All 35 features were generated to match the structure of the training data.

While this approach allowed the model to process the data, it introduced limitations as the approximated features might not perfectly reflect actual user behavior.

5.3.3 Model Performance Results

Table 5.2 Mobile Usage Dataset Performance Metrics

Metric	Value
Silhouette Score	0.3933 (Good structure)
Davies-Bouldin Index	1.0191 (Good separation)
Calinski-Harabasz Score	729.14
Cluster Distribution	Safe: 580 (58.0%), At-Risk: 420 users (42.0%)

The model achieved moderate clustering quality on the mobile usage dataset. The resulting distribution primarily favored Safe classifications, which aligns with the lower average usage of 2.20 hours per day.

5.4 Evaluation on Students Social Media Addiction Dataset (Dataset 3)

5.4.1 Dataset Overview and Demographics

The Students Social Media Addiction dataset focuses on student populations and highlights demographic usage trends.

DATASET INFORMATION:

- Original training dataset: Social Media Addiction (multi-platform)
- Test dataset: Mobile Usage Behavioral Analysis
- Test samples: 705 users
- Features used: Age, Gender, Avg Daily Usage Hours (daily)

5.4.2 Prediction

PREDICTION RESULTS:

- Safe : 0 users (0.0%) | Avg: NaN h/day
- At-Risk : 705 users (100.0%) | Avg: 4.92 h/day
- Addicted : 0 users (0.0%) | Avg: NaN h/day

5.5 Summary

This chapter provided a comprehensive evaluation of the K-Means clustering model across three separate datasets, revealing key insights into its generalization and practical use:

- Outlier Dataset: Achieved 98.78% accuracy on 82 extreme high-usage cases, showing excellent extrapolation and robust identification of severe addiction.
- Mobile Usage Dataset: Displayed moderate clustering quality (silhouette 0.3933) with appropriate adaptation to a low-usage population (58% Safe).
- Students Dataset: Successfully identified demographic-specific risk patterns (100% At-Risk) consistent with existing literature on student populations.

The evaluation identified strengths such as handling extreme cases and demographic sensitivity, alongside limitations like the requirement for complete feature sets and platform assumptions. The model serves effectively for screening and monitoring but is not intended for clinical diagnosis without further validation. Future research will explore hybrid approaches, ensemble methods, and longitudinal studies to enhance model accuracy.

Overall, the model demonstrates significant potential for real-world addiction risk assessment, particularly for high-risk individuals, emphasizing that comprehensive behavioral data is essential for peak performance.

Chapter 6

Conclusion

6.1 Introduction

This project successfully developed and implemented an AI-driven framework for the early detection of social media addiction using unsupervised machine learning techniques. The proposed system was designed to address the challenge of unlabeled behavioral data by utilizing K-Means clustering and data-driven labeling strategies. By focusing on real-world usage behavior across multiple social media platforms, the framework provides an objective and scalable solution for identifying addiction risk patterns.

The complete pipeline integrated data cleaning, feature engineering, exploratory data analysis, feature scaling, and clustering. It further employed dimensionality reduction via PCA and a dominant platform-based labeling mechanism. Experimental results demonstrate that meaningful behavioral groups can be extracted from raw usage data and transformed into interpretable addiction risk categories, ensuring the system does not rely on subjective survey-based labels.

6.2 Key Findings and Contributions

Several significant outcomes were achieved throughout this project. First, the clustering-based approach successfully stratified users into Safe, At-Risk, and Addicted behavioral groups based on usage intensity and patterns. Clustering quality metrics and PCA visualizations confirmed that the model was capable of uncovering distinct behavioral structures within the dataset.

Second, the integration of dominant platform analysis enhanced the accuracy and interpretability of

addiction classification. This hybrid strategy ensured that users with strong platform-specific dependencies were correctly identified, thereby reducing misclassification errors that often arise from aggregate usage analysis.

Third, the evaluation across multiple datasets demonstrated the robustness and adaptability of the proposed framework. The model exhibited strong performance in detecting extreme addiction cases and effectively adapted to datasets with varying usage distributions and student behavioral data. These findings validate the practical reliability of the system for real-world screening and monitoring applications.

Overall, the project presents a novel unsupervised learning-based solution that bridges the gap between behavioral analytics and digital well-being research by providing a scalable and interpretable AI-based addiction detection framework.

Final Remarks This project highlights the potential of artificial intelligence in addressing emerging digital health challenges. By combining unsupervised machine learning, behavioral feature engineering, and domain labeling strategies, the proposed system offers a practical foundation for early addiction risk identification that does not require manually labeled datasets.

The results underscore the importance of data-driven behavioral monitoring for promoting responsible technology use and digital well-being. The framework established in this research serves as a valuable reference for future academic studies, digital health applications, and policy intervention programs aimed at reducing the negative impacts of social media usage.

In conclusion, this project demonstrates that unsupervised machine learning can be effectively utilized for social media behavior analysis, marking a meaningful step toward intelligent, automated, and scalable addiction risk assessment systems.

Bibliography

[1] S. Abdel Wahed and M. Abdel Wahed. Ai-driven digital well-being: Developing machine learning model to predict and mitigate internet addiction. *LatIA*, 3:73, 2025.

[2] M. A. Mansoor and K. H. Ansari. Early detection of mental health crises through artificial intelligence-powered social media analysis: A prospective observational study. *J. Pers. Med.*, 14(9):958, 2024.

[3] A. T. B. S. Tegar, H. Hasanah, and I. Oktaviani. Predicting social media addiction using machine learning and interactive visualization with streamlit. *bit-Tech*, 8(1):778–788, 2025.

[4] J. W. Tukey. *Exploratory Data Analysis*. Addison-Wesley Publishing Company, 1977.

[5] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics. Springer, 2nd edition, 2009.

[6] D. Arthur and S. Vassilvitskii. k-means++: The advantages of careful seeding. In *Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1027–1035, 2007.

[7] P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.

[8] D. L. Davies and D. W. Bouldin. A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2):224–227, 1979.

[9] T. Calinski and J. Harabasz. A dendrite method for cluster analysis. *Communications in Statistics - Theory and Methods*, 3(1):1–27, 1974.

[10] E. Bakshy, S. Messing, and L. A. Adamic. Exposure to ideologically diverse news and opinion on facebook. *Science*, 348(6239):1130–1132, 2015.

[11] O. Hirschman. The paternity of an index. *The American Economic Review*, 54(5):761–762, 1964.

[12] G. E. P. Box, W. G. Hunter, and J. S. Hunter. *Statistics for Experimenters: Design, Innovation, and Discovery*. Wiley-Interscience, 2nd edition, 2005.

[13] J. MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 281–297, 1967.

[14] B. Mohit. Smartphone usage and behavioral dataset. Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/bhadramohit/smartphone-usage-and-behavioral-dataset.

[15] A. Singh. Student social media addiction. Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/aryansingh2000/student-social-media-addiction.